How long do two PING networks take to synchronize?

exp085 · 19 August 2026 · Draft

Abstract

This is a demonstration, not an investigation. We will search for parameters that produce a clear transition from phase drift to phase locking. We will then use one successful run to show the transition and measure its duration. It will not support a general claim about PING synchronization.

Methods

This experiment has not been run. Each method includes its planned result.

1. **Define the network.** Use SNNLANG to build two PING networks. Each has 80 excitatory cells, 20 inhibitory cells, and an independent 128-channel spike input. Start from PING parameters that produced stable rhythms in earlier work. Set their inhibitory decay times to 4 ms and 5 ms so that the uncoupled rhythms differ. Use bidirectional excitatory projections from each network to both populations in the other network. This is the primary topology because long-range excitation can recruit both parts of each local PING loop. E-to-E-only and E-to-I-only coupling remain fallback options.

   Result

   - **Axes.** None.
   - **Traces.** None.
   - **Displayed elements.** Both PING networks, their E and I populations, independent inputs, internal connections, and cross-network connections.
   - **Why.** Confirm that the implemented graph matches the design.
   - **Expectation.** Two complete PING circuits with four reciprocal AMPA projections.

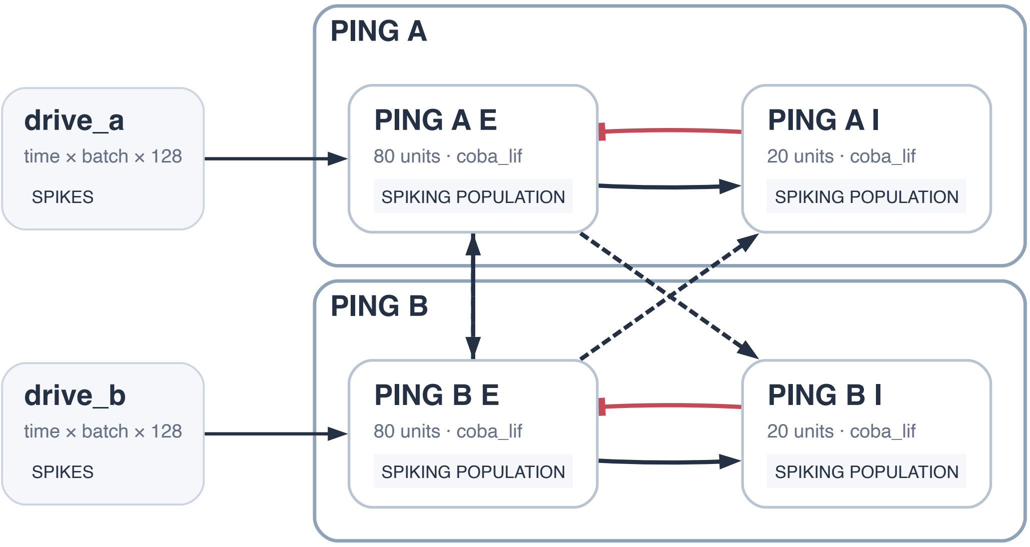

Figure 1: SNNLANG circuit view of the proposed graph. PING A and PING B each contain an excitatory and inhibitory population. Four reciprocal AMPA projections share the selected weight K after coupling.

2. **Confirm the uncoupled rhythms.** Run both networks with $K = 0$. Confirm that each produces a stable rhythm and that their wrapped phase difference repeatedly crosses the phase range.

   Result

   - **Axes.** Time before coupling on the horizontal axis; wrapped phase difference from $-\pi$ to π on the vertical axis.
   - **Trace.** Phase difference between Networks A and B.
   - **Why.** Confirm that the uncoupled rhythms drift.
   - **Expectation.** A repeating sawtooth as the faster rhythm passes the slower rhythm.

3. **Select coupling parameters.** Test candidate values of coupling weight K and coupling delay d. For each pair (K, d), switch on coupling and track the wrapped phase difference. Accept a pair if the phase difference enters a narrow band and stays there for a set hold time. Reject it if either PING rhythm stops or its population rate moves outside a stated tolerance from baseline. From the accepted pairs, choose the smallest K that gives clear locking. Record (K, d) for the demonstration.

   Result

   - **Example phase traces.**
     - **Axes.** Time after coupling on the horizontal axis; wrapped phase difference on the vertical axis.
     - **Traces.** One accepted pair and representative rejected pairs, including continued drift and a disrupted rhythm.
     - **Why.** Make the acceptance rule visible.

    - **Expectation.** The accepted trace stays in the locking band. Rejected traces drift, escape the band, or lose a valid rhythm.
- **Parameter map.**
    - **Axes.** K on the horizontal axis; d on the vertical axis.
    - **Marks.** One point per pair, coloured by accepted or rejected, with the selected pair highlighted.
    - **Why.** Record how (K, d) was chosen.
    - **Expectation.** At least one accepted region and a selected pair at its smallest successful K. This does not show robustness.

4. **Switch on coupling.** Set the selected delay d before the run. At a fixed time, change only K from zero to its selected value. Keep all other parameters unchanged. Continue until the phase difference becomes stable. Record all spikes and calculate each population firing rate.

   Result

   - **Axes.** Local time on the horizontal axis; normalized excitatory population rate on the vertical axis.
   - **Traces.** Networks A and B in three panels: before coupling, during transition, and after locking.
   - **Why.** Show how their volley timing changes.
   - **Expectation.** The peak offset drifts, adjusts, then becomes fixed.

5. **Measure the phase difference.** Use each excitatory population volley as a rhythm marker. Use these markers to calculate the phase difference $\varphi(t)$. Save the spike rasters, population rates, and volley times.

   Result

   - **Axes.** Time on the horizontal axis; population rate and phase on the vertical axes of aligned panels.
   - **Traces.** Both population rates, detected volley markers, and both phase estimates.
   - **Why.** Validate the phase measurement.
   - **Expectation.** Each phase trace follows its volley rhythm and resets at detected volleys.

6. **Measure phase locking.** The networks lock when their phase difference enters a narrow band and stays there for a set time. Report the locked phase difference. Measure t_{sync} from coupling onset to the first sustained entry into the band.

   Result

   - **Axes.** Time on the horizontal axis; phase difference $\varphi(t)$ on the vertical axis.
   - **Trace.** $\varphi(t)$, with coupling onset, the locking band, and t_{sync} marked.
   - **Why.** Measure the transition to phase locking.
   - **Expectation.** Phase drift before coupling, followed by sustained entry into the locking band.